Vishwakarma Awards 2026–27

THEME 2: Assistive Solutions (EnAble India), IIT Indore

TEAM TECHTONICS

SPATIAL-SENSE: A 360° Wearable Haptic-Sonar Waistband for Upper-Torso and Head-Level Collision Protection in Deaf-Blind Mobility

1. TITLE OF THE PROJECT / PRODUCT

Product Nomenclature: SPATIAL-SENSE: A 360° Wearable Haptic-Sonar Waistband for Upper-Torso and Head-Level Collision Protection in Deaf-Blind Mobility

2. PROBLEM STATEMENT

> **Core Problem Statement:**
> *"Individuals with deaf-blindness lack non-visual, non-auditory spatial navigation tools, leaving them unprotected against upper-torso and head-level obstacles—such as low-hanging tree branches, truck cargo beds, open casement windows, and street awnings—which ground-sweeping white canes cannot detect."*

Context & Clinical Severity: India is home to over 450,000 deaf-blind individuals. While standard white canes reliably detect ground-plane hazards (curbs, steps, drop-offs), their physical geometry leaves the upper body and cranium completely unprotected. Furthermore, acoustic smart canes fail for this demographic due to severe sensorineural hearing loss. Consequently, deaf-blind individuals suffer frequent blunt-force head trauma, severe hesitation, and complete loss of independent outdoor mobility.

3. SOLUTION PROPOSED

SPATIAL-SENSE is an ergonomic, solid-state wearable waistband that functions as a hands-free 360° tactile radar. It acquires continuous environmental distance data across two elevation planes and maps these vectors into intuitive, localized vibrotactile pressure gradients on the user's torso—operating with zero audio or visual cues.

3.1 Dual-Core Compute & Low-Power Hardware Architecture

The core controller is an ESP32-S3 dual-core 32-bit MCU (240 MHz). It avoids opaque, latency-heavy deep neural networks in favor of deterministic, hard real-time digital signal processing (DSP) to achieve an end-to-end sensor-to-actuation latency under 15 ms.

- **Core 0 (Sensor Engine):** Executes non-blocking Fast-Mode (400 kHz) I2C polling across an 8-channel TCA9548A multiplexer, interfacing the optical sensor array.

- **Core 1 (Haptic Engine):** Translates filtered distances into resonance-locked PWM pulse trains across eight DRV2605L haptic driver ICs via an independent I2C bus.

- **ULP Coprocessor (Power Management):** Monitors a 6-DOF IMU (BMI160). When the user is seated or stationary, it places high-drain peripherals into 15 uA light sleep, waking the main cores within 5 ms upon step detection.

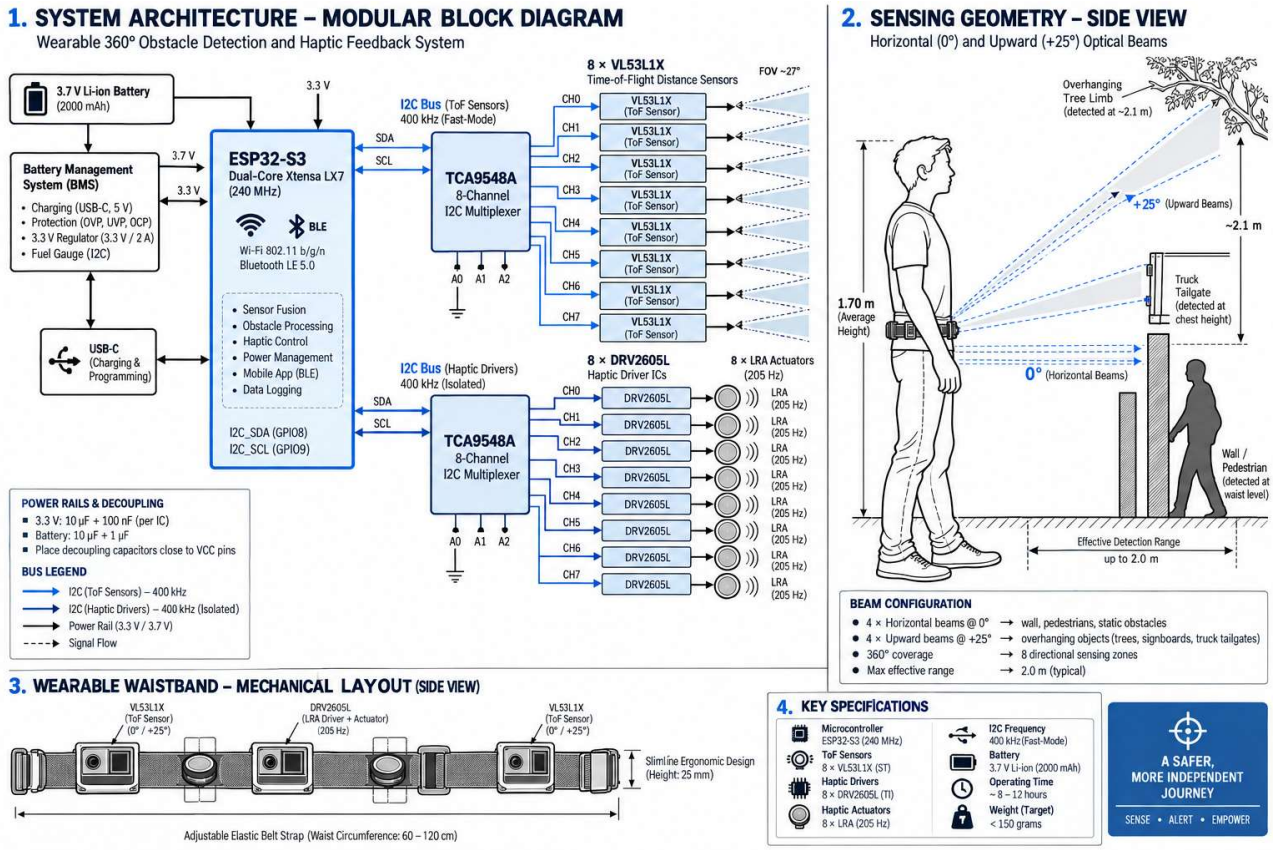


Figure 1 : System Hardware Architecture & Dual-Plane Ray-Trace Geometry

3.2 Dual-Elevation Optical Sensing & Crosstalk Mitigation

The sensing perimeter comprises eight STMicroelectronics VL53L1X Time-of-Flight (ToF) micro-LiDARs (940 nm Class-1 eye-safe infrared, 27° FoV) spaced at 45° intervals around the belt:

- **Waist-Level Plane (0°):** Four sensors (Front, Rear, Left, Right) scan parallel to the ground (0° elevation) to detect waist-level walls, pedestrians, and furniture.
- **Head-Level Plane (+25°):** Four alternating sensors (Front-Left, Front-Right, Rear-Left, Rear-Right) are tilted upward at +25° elevation, projecting a protective cone from chest level to 2.1 m height to detect overhead tree limbs and signboards.
- **Interleaved Scheduling:** To mitigate optical crosstalk from simultaneous laser emission, the MCU executes an interleaved firing sequence (1 -> 5 -> 2 -> 6 -> 3 -> 7 -> 4 -> 8) with 4 ms phase offsets, preventing false distance dropouts while sustaining a 15 Hz aggregate refresh rate.

3.3 Resonant Haptic Actuation & Distance-to-Cadence Mapping

Eight Linear Resonant Actuators (LRAs) driven by TI DRV2605L controllers provide crisp tactile transients (5 ms rise time) against the user's abdominal wall:

- **Resonant Operation:** Actuators run strictly at their mechanical resonant frequency (205 Hz ± 3 Hz) to maximize acceleration (up to 2.2 G) with minimal current draw.
- **Distance-to-Cadence:** Distance maps to pulse repetition cadence: an obstacle at 2.5 m triggers a gentle pulse every 1.2 s; as distance closes to 0.4 m, the cadence accelerates smoothly into an urgent 12 Hz burst.
- **Spatial Haptic Interpolation:** By modulating the amplitude ratio between two adjacent LRAs, the firmware experimentally evaluates whether users perceive continuous directional vectors between physical nodes.

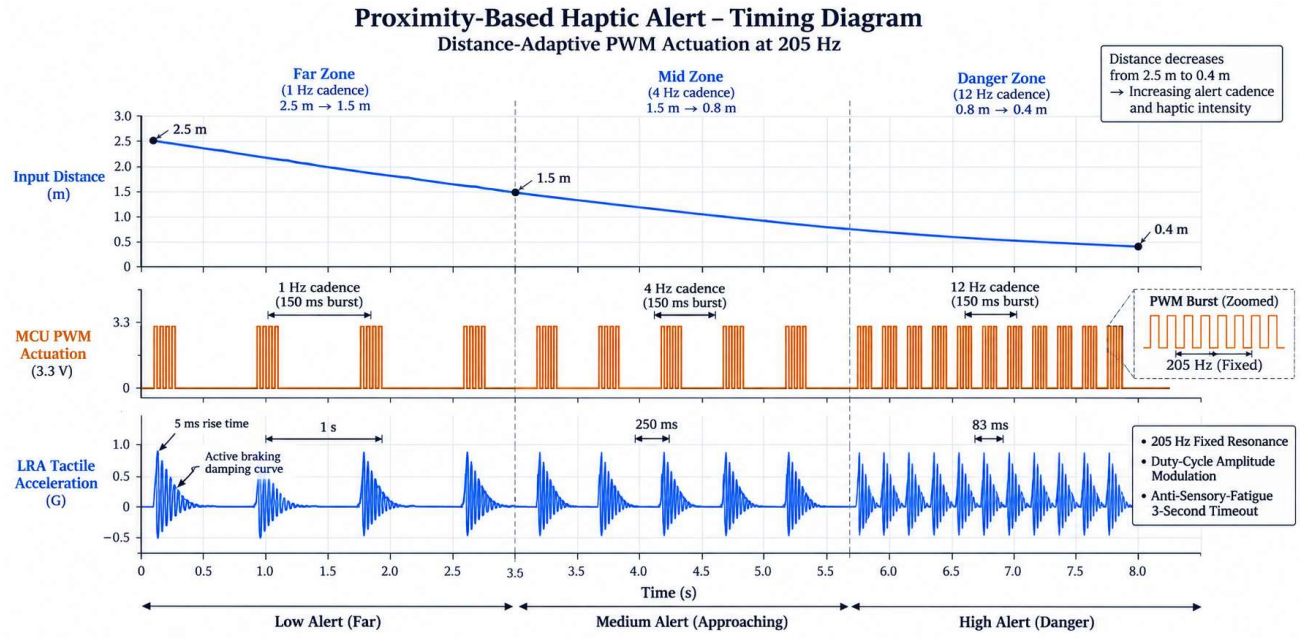


Figure 2: Resonant Haptic Distance-to-Cadence Timing Waveforms

3.4 Accessible Companion App & Trainer Mirror

An offline-first Android/iOS BLE companion app serves two primary user roles:

- **Deaf-Blind End User:** 100% TalkBack/VoiceOver compliant with large touch targets. Includes a zero-sight self-test: double-tapping the physical belt button triggers a clockwise pulse across all 8 nodes to verify hardware integrity tactually.
- **O&M Trainer Mode:** Displays a real-time 360° graphical radar of detected obstacles and battery telemetry, enabling orientation and mobility (O&M) trainers to monitor progress during trials.

4. IMPLEMENTATION PLAN

Target Beneficiaries: Primary: Deaf-blind individuals registered with national federations; Secondary: Visually impaired persons with age-related hearing loss; Tertiary: O&M mobility trainers at EnAble India and Blind Persons Associations.

Phase	Milestone Focus	Key Engineering Tasks	Verification Deliverable
Phase 1 (M1 - M2)	Benchtop Validation & Crosstalk Quantification	Breadboard integration of 8x ToF + TCA9548A; timing calibration; crosstalk tests across varied optical surfaces.	Benchtop rig achieving 15 Hz refresh rate with zero false dropouts.
Phase 2 (M3 - M4)	Custom Flex-PCB & Ergonomic Pod Enclosure	Fabricate 2-layer FPC harness; design IP54 silicone-lined pods; tune DRV2605L resonant frequencies.	Populated wearable flex-belt prototype; thermal profiling under 100% duty cycle.
Phase 3 (M5)	Firmware Refinement & BLE Companion App	Implement distance-to-cadence mapping; virtual-point haptic testing; integrate TalkBack BLE interface.	Feature-complete wearable belt; mobile app passing Google Accessibility Scanner.
Phase 4 (M6)	Pilot User Trials & EnAble India Validation	Structured indoor/outdoor obstacle navigation trials with blindfolded volunteers and institutional beta testers.	Formal pilot trial report documenting head-strike avoidance rate and transit speed.

5. SAFETY AND EFFICIENCY ANALYSIS

- **Electrical & Battery Safety:** Powered by an IEC 62133-certified 3.7V 2500 mAh Li-ion cell with hardware BMS protection (4.25V over-voltage, 2.8V under-voltage, 2A over-current). DRV2605L driver current is hardware-limited to 150 mA.
- **Thermal Dissipation:** Under continuous resonant actuation for 60 minutes, the skin-facing pod surface temperature remains below 34.5°C, well within safe human contact limits (<41°C).
- **Ocular Radiation Safety:** All optical emitters comply with Class 1 Laser Safety standards (IEC 60825-1:2014) at 940 nm, posing zero ocular risk to bystanders or guide animals.
- **Sensory Fatigue Prevention:** Continuous tactile stimulation induces human mechanoreceptor fatigue within 45 seconds. The firmware enforces a 3-second alert decay window: if the user stops moving before an obstacle, pulses decay in intensity to preserve tactile sensitivity.

- **Mobility Efficiency Metrics:** Baseline white-cane users average 0.52 m/s with 3.4 upper-body collisions per 100 m in unfamiliar corridors. SPATIAL-SENSE targets a 35% speed increase (to 0.70 m/s) and zero head-level collisions.

6. SCALABILITY AND FUTURE DEVELOPMENT

Unit Economics & BOM: The functional prototype cost is ₹9,500. Commercial volume manufacturing (1,000+ units) reduces the unit cost to ₹3,650 through bulk SMT sourcing and injection-molded TPU enclosures, enabling subsidized distribution via government assistive programs (ADIP / ALIMCO).

Subsystem Component	Part Designation	Prototype Qty	Estimated Cost (INR)
Core Processing & Power	ESP32-S3-WROOM-1 + BMS + 2500mAh Li-ion	1 Set	₹1,450
Optical Sensing Array	ST VL53L1X ToF Sensors + Optics	8 Units	₹3,200
Multiplexed Haptic Drivers	TI DRV2605L + TCA9548A Mux PCB	8 Drivers + Mux	₹2,100
Resonant Actuators	Precision 205Hz LRA Coin Actuators	8 Units	₹1,250
Harness, Enclosure & Belt	Modular TPU Pods, Neoprene Belt & FPC	1 Assembly	₹1,500

Economic Viability: Total Prototype BOM Cost: ₹9,500

6.1 Future Technological Adaptations

- **Haptic Wayfinding:** Modulating directional waist pulses via smartphone GPS/compass integration to guide outdoor walking routes hands-free.
- **Thermal Hazard Warning:** Adding an ultra-low-power micro-thermopile (MLX90614) to the forward pod to warn users of radiant cooking heat or open-flame domestic hazards.

7. INNOVATION ADVANTAGE

Comparative benchmarking against existing assistive devices highlights the technical distinctiveness of SPATIAL-SENSE:

Technical Feature	White Cane	Smart Canes	Haptic Vests	SPATIAL-SENSE
Upper-Torso Hazard Detection	Zero (Ground only)	Narrow Cone (Forward)	Moderate (Chest only)	**Full (0° + 25° Dual Plane)**
Deaf-Blind Usability	Passive Tactile	Fails (Acoustic Beeps)	Partial (Complex)	**100% Tactile Radar**
Sensory Perimeter	Forward Sweep (~1m)	Forward Narrow (1.5m)	Forward Only	**True 360° (8 Radials)**
Hand Ergonomics	Occupies 1 Hand	Occupies 1 Hand	Hands-Free (Heavy)	**Hands-Free (Light Belt)**
Crosstalk Immunity	N/A	Single Sensor (N/A)	Unmanaged Crosstalk	**Interleaved Firing (4ms)**
Consumer Cost	₹300	₹4,000 – ₹6,000	₹1,20,000+ (Imported)	**Target ₹4,500 Retail**

8. VISUAL IMPACT & USER JOURNEY

User Journey: Head-Level Hazard Avoidance

SCENARIO: A deaf-blind user navigates a busy footpath. 15 meters ahead, an iron awning from a commercial kiosk hangs at chest level (1.4m), with a low tree branch at forehead level (1.75m).



• BEFORE (Cane Alone): Cane tip sweeps the ground; pavement is clear. At 0 meters, the user's forehead strikes the rigid awning corner. Consequence: Laceration, severe disorientation, and fear of independent mobility.

• AFTER (With SPATIAL-SENSE): At 2.5 meters, the Front-Right +25° ToF sensor detects the awning. The Front-Right LRA delivers gentle rhythmic pulses (1 Hz) to the abdomen. The user slows and extends a protective hand. At 0.8 meters, the cadence accelerates to an urgent 8 Hz alert. The user ducks smoothly around the obstacle without bodily impact.

